A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 14

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Mars has low mass so weak gravity and could not retain a dense atmosphere **1** Very low atmospheric pressure means water would boil / evaporate away even at low temperatures **1** Greater distance from the Sun gives low insolation and a mean temperature of -60 °C so water is frozen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Mass great enough for gravity to retain an atmosphere containing the elements needed (C, H, O, N as CO_2, methane, ammonia, water vapour) and to maintain pressure so liquid water does not boil **1** Distance from the Sun (the 'Goldilocks zone') gives insolation producing temperatures at which water is liquid (0-35 °C over most of the surface) and proteins are not denatured **1** Too close (Venus) - too hot – too far (Mars) - too cold **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** The molten core produces a magnetosphere that deflects the solar wind / charged particles, preventing biologically damaging radiation reaching the surface **1** The 24-hour rotation limits heating by day and cooling by night, reducing temperature extremes (Venus' slow rotation gives extreme conditions) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** The tilted axis causes seasonal variation in insolation as the Earth orbits the Sun **1** Producing seasons (temperature, day length) to which species' life cycles (breeding, migration, dormancy) are adapted – the Moon's gravity keeps the tilt stable and creates tides **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
02.1	1 mark for the point below. **1** Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Zebu tolerate heat but have a low milk yield – Ayrshire have a high milk yield but are not heat tolerant **1** Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	2
02.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Replacement of valuable animals that die with genetically identical individuals **1** Replacement of herds culled during a disease outbreak **1** Production of large numbers of individuals with desirable characteristics **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
03.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Copying the structural / functional adaptations of living organisms in the design of manufactured items **1** Aircraft wingtip fins copied from splayed wingtip feathers of soaring birds, reducing turbulence and fuel use **1** Humpback whale flipper tubercles copied on ship rudders and wind turbine blades (tighter turns, rotation at lower wind speeds) **1** Shark skin ridges reducing friction / bacterial adhesion (surface coatings, hospital surfaces) **1** Termite mound convection copied for natural ventilation of buildings – hollow strutted bird bones in lightweight roofs – gecko toe adhesion – burr hooks - Velcro – lotus leaf self-cleaning glass – spider silk for airbags / body armour – bat / dolphin echolocation - ultrasound scanners **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	4

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Plants produce toxic chemicals (eg alkaloids) to defend themselves against herbivores – in controlled doses these affect human physiology **1 Examples:** morphine / codeine from poppies – quinine from cinchona bark (malaria) – aspirin from willow bark – taxol from yew (cancers) – diosgenin from Mexican yam (contraceptives, cortisone) **1 Animal examples accepted:** AZT from a marine sponge (HIV), shark liver lipids (tumours) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Only a very small proportion of species have been studied for medicinal substances so unknown species may contain valuable chemicals **1** Conserving the habitat protects all its species, including those not yet investigated (and their interdependent relationships) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	1 mark for the point below. **1** Squid giant nerve cells - nerve function / heart disease, stroke – marsupials (kangaroo, wallaby pouch) - embryo development – hippo skin secretion - sunscreen / burns – sea urchin embryos - teratogen testing – armadillo - leprosy – marine sponge proteins - transplant rejection **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-15; AQA spec 3.1.2.1.1-3.1.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 4 points below. **1** Very small gene pool so individuals are closely related and inbreeding is likely **1** Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low **1** A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) **1** Exchange of individuals / semen between zoos – avoid hybridisation with related species / subspecies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
04.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided **1 Population interactions:** mixing breeding and non-breeding individuals causes conflict – some need a choice of partners, others pair for life **1 Breeding habitat needed:** courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) **1** Stress / disturbance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
04.3	**1 mark each** 1 mark for each of the 3 points below. **1 AI:** semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet – avoids dangerous mating (elephants, rhinos) and partner rejection **1 Embryo transfer:** a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) **1** So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Drainage makes soils aerobic and drier, increasing grass growth but wetland plants and the animals they support (waders, amphibians, invertebrates) cannot survive **2** Fertiliser increases nutrient availability so vigorous species (grasses, nettles) grow taller and out-compete smaller wildflowers for light – diversity falls **2** Hedgerow removal destroys habitat / corridors and nesting sites – loss of natural pest predator habitats increases pest damage – larger fields increase wind erosion – genetic contamination and introduced species also accepted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Long-continued traditional farming has created plagioclimax habitats colonised by many species: hay meadows, chalk grassland, grazed heath / moorland, hedgerows **1** These communities depend on the farming continuing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** CO2 from fossil fuel use in machinery and embodied energy of fertiliser (Haber process) and machinery – ploughing increases soil respiration **1** Methane from livestock intestines and rice paddies – nitrous oxide from manure and nitrogen fertilisers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-330; AQA spec 3.5.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Species:** a group of closely related organisms that naturally interbreed to produce fertile offspring and are reproductively isolated from other groups (Passer domesticus, house sparrow) – the only 'real' taxon **2 Niche:** the role a species plays in its habitat - how it uses resources and relates to other species (tawny owl: nocturnal carnivore nesting in tree holes eating small mammals) – two species cannot share exactly the same niche in one habitat **2 Community:** all the populations of all species (plants, animals, fungi, bacteria, protists, archaea) living in an area **2 Biome:** a large geographical region with a specific climate and characteristic community, including all areas where it occurs (all tropical rainforests, tundra, savannah) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **Niche** → the role a species plays and the conditions it needs	8
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** An ecosystem is the biotic community plus abiotic factors of one area, relatively self-contained (a particular coral reef) **1** A biome includes all separate ecosystems with the same climate and community type (all coral reefs) – each reef is a different ecosystem within the same biome **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biotic** → living factors / effects of other organisms • **abiotic** → non-living / physical factors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 82-85; AQA spec 3.1.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 4 points below. **1 Energy:** low energy-density renewable (solar) / non-renewable fossil fuels **1 Materials:** cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion **1 Wastes:** biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate **1 Temperature:** low / high (Haber process, incineration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time • **incineration** → burning waste, usually with energy recovery **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) **1** They are re-used in cycles so there is no depletion of raw materials or build-up of wastes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**2 marks** 2 marks for the point below; award fewer where it is only partly made. **2** Use often disperses the minerals (eg in products spread over the world / corrosion) **OR** produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** They cannot easily be removed from end-of-life appliances and are not biodegradable **1** Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1 Moral belief:** protecting the environment is the 'right' thing to do regardless of benefit to people **1 Enlightened self-interest:** protecting the environment is done because it causes fewer difficulties (for people) than not doing so **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Total impact = population size x the individual (per capita) impact of each person **1** Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
08.3	**Any 1 of 3** 1 mark each for any 1 of the 3 points below, in any order, maximum 1. **1** Slash and burn agriculture **1** Fishing **1** Discharge of sewage / biodegradable pollutants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **biodegradable** → broken down by decomposers **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** Each regional population has its own unique gene pool / DNA profile **1** DNA from seized timber or ivory is compared with the database of regional populations **1** This identifies where the material came from, showing whether it was legally harvested and where poaching occurs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Radio / GPS transmitters use radio waves - work in air and fresh water, giving positions (GPS stores or transmits location) **1** Radio waves do not travel through seawater so acoustic transmitters (sound) are used for marine animals, giving real-time positions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2** Drones / ROVs, AUVs (eg Autosub), balloons, aircraft, satellites, animals (tags) **up to 2 Drones:** cheap, low disturbance, high-resolution images of areas hard to reach (cliffs, canopy, ice), repeated surveys – infrared cameras detect animals at night **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Cameras record automatically (visible light by day, infrared at night) without researchers being present **1** So animals behave naturally and nests are not deserted – monitoring continues 24 hours **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-406; AQA spec 3.7.2.3.	
10.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Zooxanthellae are photosynthetic algae living inside the polyp cells **1** The algae absorb CO_2 and excretory wastes from the polyp for photosynthesis and growth (efficient nutrient recycling) **1** They supply the polyp with oxygen and glucose - over 95% of its food energy **1** In return the algae gain a protected habitat and nutrients they could not absorb from the nutrient-deficient water **1** Polyps also feed on plankton captured by cilia and nematocysts **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	4
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Corals deposit a skeleton of calcium carbonate (calcite) **1** Each carbonate molecule requires the absorption of a molecule of carbon dioxide, so reefs are a carbon sink **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
10.3	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Importance:** cover 1% of ocean area but support over a quarter of marine species – fisheries (fish, lobsters, conch) – medicines (AZT, Ara-C, dolastatin, secosteroids, bryostatin, trabectedin, bone grafts) – climate control as carbon sink – erosion protection of low-lying islands – tourism income • **Threats:** physical damage (anchors, divers, litter, coral extraction), collection of ornaments (CITES Appendix II), sedimentation from rivers and coastal development, pollution (oil, nutrients causing algal shading, sewage, pesticides, sunscreen), ocean acidification, bleaching from warming, overfishing (giant triton / crown of thorns starfish), introduced species (red lionfish), loss of mangroves and seagrass nurseries • **Conservation:** marine protected areas / national parks (Great Barrier Reef, Belize, Chagos), mooring buoys, diver education, No Take Zones which can increase catches elsewhere, sustainable ecotourism and fishing, protection of mangroves and seagrass, erosion and pollution control, planning control of ports and resorts • **Fishing controls:** closed seasons, minimum and maximum catchable sizes, bans on dynamite, trawling and small-mesh nets • **Evaluation:** local measures cannot control global threats (warming, acidification) which need international action – enforcement in remote areas is difficult – conflicts with local livelihoods – large vs small NTZs • Justified conclusion **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **CITES** → the treaty that controls international trade in endangered species • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	
	Model answer - Level 3 (9 marks)	
	Tropical coral reefs cover about one per cent of the oceans yet support over a quarter of marine species, a diversity built on the symbiosis between polyps and zooxanthellae that recycles nutrients in nutrient-poor water. They matter to people for fisheries of fish, lobsters and conch; for medicines such as AZT, Ara-C, dolastatin, secosteroids, bryostatin and trabectedin discovered in reef organisms; as a carbon sink, since every carbonate unit in coral skeleton absorbs a molecule of carbon dioxide; for dissipating wave energy that would erode low-lying islands; and for tourism income.	
	They are also unusually vulnerable because they need narrow abiotic conditions. Anchors, divers, litter and coral extraction kill polyps; ornaments are collected; sediment from rivers and coastal development smothers them; oil, sewage and nutrients cause algal shading; ocean acidification hinders skeleton formation; warming causes bleaching; overfishing of giant tritons lets crown-of-thorns starfish destroy reefs; introduced lionfish eat native fish; and loss of mangroves and seagrass removes the nurseries and sediment traps reefs depend on.	
	Marine protected areas such as the Great Barrier Reef, Belize and Chagos combine mooring buoys, diver education, no-take zones that raise catches outside them, sustainable tourism and protection of linked habitats, while fishing controls set closed seasons, minimum and maximum sizes and ban dynamite, trawling and small mesh. These work well against local threats where they are enforced, and no-take zones have measurable benefits, but they cannot touch warming and acidification, which need international action, enforcement in remote areas is weak, and restrictions conflict with local livelihoods. Local conservation buys time; only global climate action can secure the reefs.	
	Weaker answer - Level 1 (3 marks)	
	Coral reefs are important because lots of fish live there and people catch them for food. Tourists visit reefs to dive and this brings money. Reefs also protect the coast from waves. Reefs are threatened by climate change which causes bleaching, and by pollution and damage from boats. They are conserved by making marine protected areas where fishing is banned. This helps but climate change is still a problem.	
	What makes the difference	
	• Level 1 has the right headings but Level 3 explains each (why reefs are diverse, why they are a carbon sink, how each threat acts).	
	• Level 3 uses named reserves and specific management measures; Level 1 gives one generic measure.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation separates local from global threats and reaches a nuanced judgement; Level 1 states that climate change is 'still a problem'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	
11.1	**Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Definition of sustainability (Brundtland) – population x per capita impact – dynamic equilibria, feedback, tipping points, diversity • **Principles of natural systems:** renewable low-density energy, low temperatures, material cycles, non-toxic wastes – contrast with linear human systems • **Circular economy principles:** cycling materials, renewable energy, supporting ecosystems, separation of technical and biological materials, diversity, connected systems, design for extended use and re-use, optimum production, new technologies • Application to land use, water, minerals, waste, pollution, energy, food with examples (Kalundborg, Eigg, Ol Pejeta, Rio Bravo, EU initiatives, DC grids) • **Measuring progress:** ecological footprint, biocapacity, global hectares, Earth Overshoot Day, carbon footprint, Living Planet Index • **Barriers:** economics, competition, consumer behaviour, need for international agreement, existing infrastructure • Judgement on the extent achievable	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-389; AQA spec 3.6.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Dynamic equilibria and negative feedback in climate, hydrological cycle and populations – positive feedback and tipping points caused by humans • **Energy:** solar-driven, low energy density, low temperature (photosynthesis, enzymes, nitrogen fixation vs Haber process) – renewable energy, heat pumps, bioleaching, fuel cells • **Materials:** cycles, few abundant elements, biodegradable non-toxic wastes vs fossil fuels and dispersed minerals – recycling, Cradle to Cradle, industrial symbiosis • **Diversity and stability:** rainforests / reefs vs agroecosystems – diverse energy mixes • Optimum vs maximum production – supporting ecosystems in agriculture, forestry, fisheries, urban design • Case studies and metrics – limits and judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-379; AQA spec 3.6.1-3.6.4.	

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